Towards Monosemanticity: Decomposing Language Model Representations via Sparse Dictionary Learning

Ruth Sharon Raju

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1 Introduction

1.1 Motivation

The future of AI safety depends on the decisions we make today. These decisions require us to study the large language models of today in great detail. A good starting point is learning how they internally represent meaning. The field of mechanistic interpretability research offers us this opportunity, allowing us to break down neural networks into smaller components that are better understood than the whole. However, their internal representations are difficult to interpret because individual neurons are often polysemantic, responding to multiple unrelated concepts. This makes neuron-level analysis an unreliable unit for understanding model behavior. In this project, I explore the hypothesis that language models instead encode meaning in higher-dimensional directions of activation space. By applying sparse dictionary learning to model activations, I aim to recover monosemantic features that correspond to coherent concepts. This ensures a more interpretable inner view of language models and it can further assist in debugging, alignment, and scientific insight.

1.2 Problem Statement

Let

$$h^{(l)}(x) \in R^d \tag{1}$$

denote the activation vector at layer l for input token x. Given a corpus, we collect activations

$$\mathcal{H} = \{h_i\}_{i=1}^{N}, \quad h_i \in R^d. \tag{2}$$

Since individual neurons (coordinates of $h^{(l)}(x)$) in large language models are often *polysemantic*, the problem addressed is finding a representation of the activation space that is more interpretable than individual neurons. The goal is to obtain a higher-dimensional representation with clear semantic interpretation.

2 Foundations of Interpretability

Recent work on interpretability was realized through a mechanistic perspective, where neural networks are reverse-engineered by identifying the internal computations and structures that affect model behavior. [1] A central challenge motivating mechanistic interpretability is the problem of *superposition* [2]. Neural networks have limited representational capacity relative to the number of concepts they must encode, models often store multiple features in overlapping directions of activation space, leading to polysemanticity. Early progress in resolving superposition came from computational structures like induction heads, in-context learning mechanisms, and transformer circuits. [3] However, this analysis did not by itself scale to the full diversity of features. Further analyses then revealed how composition and superposition arise naturally during training and why disentangling features is difficult in high-dimensional systems.

The long-term goal is to also develop scalable methods for understanding models as they grow larger and more capable. [4] Sparse dictionary learning through sparse autoencoders is a great approach in

this direction. By learning a set of sparse features that reconstruct model activations, separate latent directions are allocated to distinct concepts.

3 Models, Methods, and Algorithms

3.1 Base Model and Corpus

The work starts by analyzing the internal representations of a pretrained GPT-2 model. Let f denote the transformer and let $h^{(l)}(x) \in R^d$ denote the hidden activation vector associated with input token x at transformer block l. Throughout, $d = 512$.

For the dataset, I curated a physics-heavy corpus in order to suggest a theme to the model to make up for the inability to implement this experiment at the original scale. The corpus consists of 82,254 non empty lines or 758,635 characters. Each line is tokenized using the GPT-2 tokenizer with padding and truncation enabled.

3.2 Activation Extraction via Forward Hooks

The goal is to learn a feature dictionary over the activation space of a specific internal component. In the implementation, activations are extracted from the MLP output projection of a chosen transformer block, i.e., the module `model.h[l].mlp.c_proj`. Concretely, for a token sequence with length L, the hooked module produces a tensor of activations

$$A \in R^{B \times L \times d}, \tag{3}$$

where B is the batch size. Each valid token position is treated as one data point for dictionary learning by flattening across batch and sequence dimensions.

Since batches are padded, padding positions are filtered out using the attention mask. Let $m \in \{0, 1\}^{B \times L}$ be the attention mask; the final activation dataset is

$$\mathcal{H} = \{a_i\}_{i=1}^{N}, \quad a_i \in R^d, \tag{4}$$

where N counts only positions with $m = 1$. To control computational cost, we cap the number of collected token activations at a fixed budget $N \leq T$ (with $T = 400{,}000$ in our experiments).

3.3 Token-Context Alignment for Feature Interpretation

For each retained activation vector a_i, (i) the decoded token string and (ii) a short decoded context window around that token are stored. Specifically, for a token at position t, I decoded a context window of size ± 5 tokens:

$$\text{context}(t) = \text{decode}(x_{t-5:t+5}). \tag{5}$$

This yields a list of records $\{(\text{token}_i, \text{context}_i)\}_{i=1}^{N}$ that is indexed identically to $\{a_i\}_{i=1}^{N}$.

3.4 Sparse Autoencoder for Dictionary Learning

The sparse autoencoder maps an activation $a \in R^d$ to a latent feature vector $z \in R^k$:

$$z = \phi(W_e a + b_e), \tag{6}$$

where $W_e \in R^{k \times d}$, $b_e \in R^k$, and ϕ is a nonlinearity enforcing non-negativity (ReLU in our implementation). The decoder reconstructs the activation as

$$\hat{a} = W_d z + b_d, \tag{7}$$

with $W_d \in R^{d \times k}$ and $b_d \in R^d$. A latent space with $k = 4096$, corresponding to an $8\times$ expansion relative to $d = 512$, is used.

3.5 Training Objective and Optimization

The SAE is trained to minimize reconstruction error subject to sparsity in the latent code. For each activation vector a_i, the loss is

$$\mathcal{L}(a_i) = \|a_i - \hat{a}_i\|_2^2 + \lambda \|z_i\|_1, \tag{8}$$

where $\lambda > 0$ controls the sparsity–fidelity trade-off. Optimization is performed using mini-batch gradient descent over the activation dataset $\mathcal{H}$.

3.6 Ranking Tokens for a Learned Feature

After training, learned features are interpreted by identifying the tokens that most strongly activate a given feature. Let $z_{i,j}$ denote the activation of feature j on token i. For each feature j, we rank tokens by $z_{i,j}$ and inspect the top-K items:

$$\text{TopK}(j) = \arg \max_{i \in \{1,\ldots,N\}}^{K} z_{i,j}. \tag{9}$$

The decoded token and its associated context are recorded for each selected index, enabling qualitative assessment of whether feature j corresponds to a coherent semantic pattern.

4 Results and Interpretation

Feature ID	Top-Activating Tokens	Interpreted Concept
8	`black hole`, `event horizon`, `singularity`, `collapse`	Black hole physics
17	`integral`, `derivative`, `matrix`, `eigenvalue`	Math notation
42	`energy`, `entropy`, `thermodynamic`, `system`	Physical systems
109	`if`, `then`, `thus`, `therefore`	Logical structure
256	`quantum`, `wavefunction`	Quantum mechanics

Table 1: **Table 4.1 Interpretation of features based on top-activating tokens**

Table 4.1 shows that several learned latent features correspond to coherent and domain-specific concepts. The top-activating tokens for each feature exhibit clear semantic alignment which is evidence that the sparse autoencoder has identified meaningful directions in the activation space.

Feature ID	Activation Density	Mean Activation	Max Activation
8	0.21%	0.205	1.80
17	0.34%	0.182	1.62
42	0.18%	0.224	1.91
109	0.41%	0.167	1.48
256	0.27%	0.198	1.73

Table 2: **Table 4.2 Summary statistics for selected features**

Table 4.2 contains the activation statistics for the same set of features. Low activation densities across features indicate sparse usage, while relatively high maximum activation values suggest strong selectivity.

5 Advantages and Limitations

5.1 Advantages

Large language models are trained almost entirely through text, and their internal representations are thus shaped primarily by textual regularities and abstractions. Feature-based interpretability methods that operate directly on token-level activations align well with this mode of learning.

GPT-2 was used as the base model as it offered activation extraction without excess computational cost, and despite its smaller scale, allowed for the scaling down of the implementation without inconvenience.

5.2 Limitations

A fundamental limitation of monosemanticity is that the presence of interpretable features does not guarantee reliable insight into a model's internal objectives. Moreover, monosemantic representations are learned with respect to observed activations and training data distributions. Features that appear benign or coherent under typical inputs may behave differently under distributional shift or adversarial prompting. Additionally, the assumption that internal representations can be cleanly decomposed into sparse, human-aligned features may not hold uniformly across all layers or models. [5]

6 Conclusion

This work explores sparse autoencoder–based dictionary learning as a tool formechanistic interpretability in transformer language models. By applying this method to GPT-2 activations, it is possible to recover interpretable, sparse features that correspond to coherent semantic concepts.

These findings support the broader view that language models encode meaning in distributed activation patterns rather than individual neurons. While the present experiments are limited in scale, they reproduce key aspects of prior work. Future work could extend this approach to multiple layers, larger datasets, cross-domain corpora,and even automated methods for feature labeling and comparison.

References

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